EDGE INTELLIGENCE – LAB TASK-2

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CODE:

```
import tensorflow as tf
from tensorflow.keras.datasets import mnist
import matplotlib.pyplot as plt
from tensorflow.keras import layers, models
```

```
(x_train, y_train), (x_test, y_test) = mnist.load_data()
```

```
Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/mnist.npz
11490434/11490434 ━━━━━━━━━━━━━━━━━━━━ 0s 0us/step
```

```
x_train = x_train.astype("float32") / 255.0
x_test = x_test.astype("float32") / 255.0
```

```
x_train = x_train.reshape(-1, 28, 28, 1)
x_test = x_test.reshape(-1, 28, 28, 1)
```

```
model = models.Sequential([
    layers.Input(shape=(28, 28, 1)),
    layers.Conv2D(32, (3, 3), activation='relu'),
    layers.MaxPooling2D((2, 2)),
    layers.Conv2D(64, (3, 3), activation='relu'),
    layers.MaxPooling2D((2, 2)),
    layers.Flatten(),
    layers.Dense(64, activation='relu'),
    layers.Dense(10, activation='softmax')
])
```

```
model.compile(optimizer='adam',
              loss='sparse_categorical_crossentropy',
              metrics=['accuracy'])
```

```
history = model.fit(x_train, y_train, epochs=10,
                    validation_data=(x_test, y_test),
                    verbose=1)
```

```
Epoch 1/10
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 32s 17ms/step - accuracy: 0.9991 - loss: 0.0027 - val_accuracy: 0.9903 - val_loss: 0.0579
Epoch 2/10
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 41s 17ms/step - accuracy: 0.9983 - loss: 0.0051 - val_accuracy: 0.9912 - val_loss: 0.0493
Epoch 3/10
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 33s 17ms/step - accuracy: 0.9992 - loss: 0.0026 - val_accuracy: 0.9918 - val_loss: 0.0454
Epoch 4/10
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 33s 17ms/step - accuracy: 0.9990 - loss: 0.0028 - val_accuracy: 0.9902 - val_loss: 0.0539
Epoch 5/10
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 41s 18ms/step - accuracy: 0.9993 - loss: 0.0018 - val_accuracy: 0.9891 - val_loss: 0.0709
Epoch 6/10
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 31s 16ms/step - accuracy: 0.9990 - loss: 0.0036 - val_accuracy: 0.9912 - val_loss: 0.0585
Epoch 7/10
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 42s 17ms/step - accuracy: 0.9990 - loss: 0.0038 - val_accuracy: 0.9906 - val_loss: 0.0599
Epoch 8/10
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 32s 17ms/step - accuracy: 0.9991 - loss: 0.0029 - val_accuracy: 0.9916 - val_loss: 0.0585
Epoch 9/10
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 31s 17ms/step - accuracy: 0.9994 - loss: 0.0025 - val_accuracy: 0.9921 - val_loss: 0.0491
Epoch 10/10
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 32s 17ms/step - accuracy: 0.9996 - loss: 0.0011 - val_accuracy: 0.9911 - val_loss: 0.0550
```

```
test_loss, test_acc = model.evaluate(x_test, y_test, verbose=2)
print(f'\nTest accuracy: {test_acc:.4f}')
```

```
313/313 - 2s - 6ms/step - accuracy: 0.9911 - loss: 0.0550

Test accuracy: 0.9911
```

```
model.save('mnist_cnn_model.keras')

# Load the model back
from tensorflow.keras.models import load_model
loaded_model = load_model('mnist_cnn_model.keras')
```

```
import pickle

# Save model using pickle
with open('mnist_model.pkl', 'wb') as f:
    pickle.dump(model, f)
```

+ Code + Text

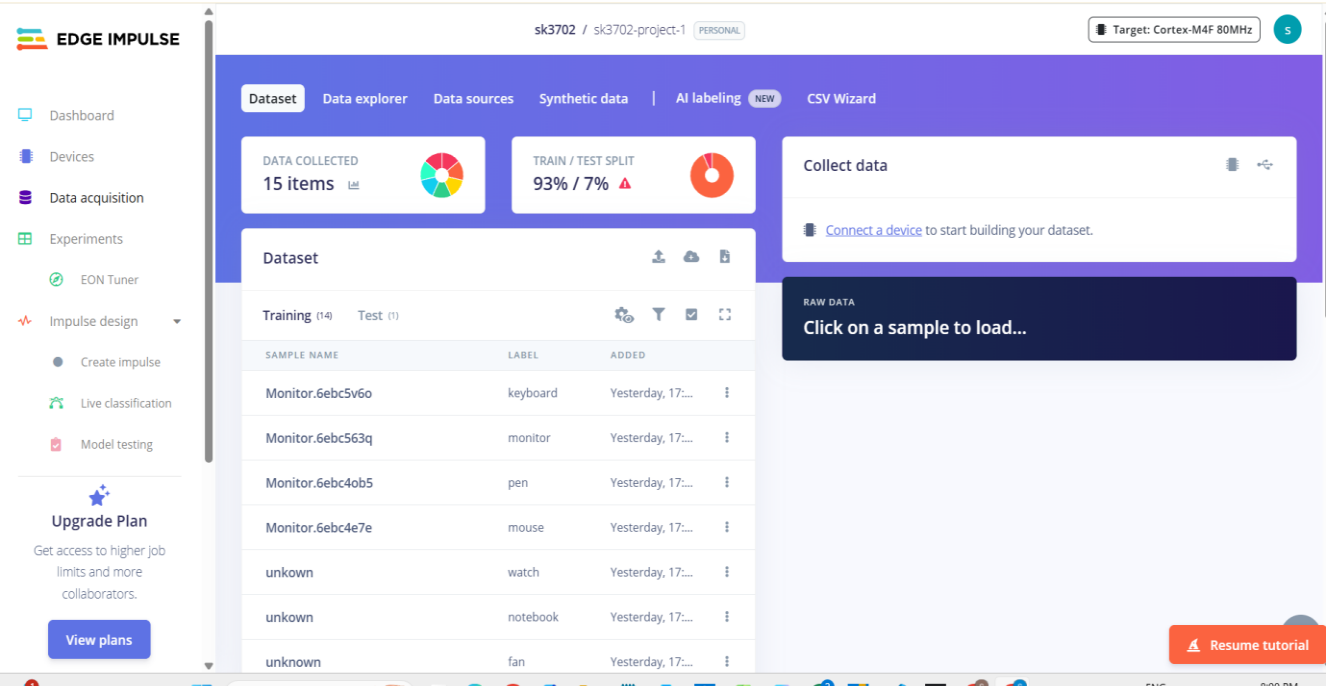
```
# Load model using pickle
with open('mnist_model.pkl', 'rb') as f:
    pickled_model = pickle.load(f)
```

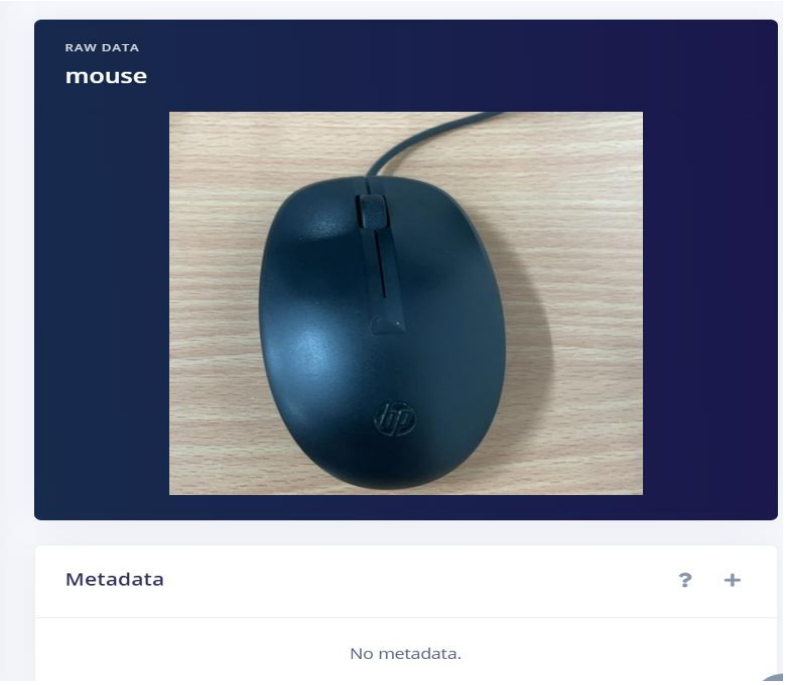
```
model.summary()
```

Model: "sequential"

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 26, 26, 32)	320
max_pooling2d (MaxPooling2D)	(None, 13, 13, 32)	0
conv2d_1 (Conv2D)	(None, 11, 11, 64)	18,496
max_pooling2d_1 (MaxPooling2D)	(None, 5, 5, 64)	0
flatten (Flatten)	(None, 1600)	0
dense (Dense)	(None, 64)	102,464
dense_1 (Dense)	(None, 10)	650

EDGE IMPULSE:





RAW DATA

fan

Metadata

RAW DATA

keyboard

Metadata

RAW DATA

pen

Metadata

RAW DATA

watch

Metadata

EDGE IMPULSE

sk3702 / sk3702-project-1 PERSONAL

Target: Cortex-M4F 80MHz

- Dashboard
- Devices
- Data acquisition
- Experiments
  - EON Tuner
- Impulse design
  - Create impulse
  - Image
  - Classifier

Upgrade Plan

Get access to higher job limits and more collaborators.

Neural Network settings

Training settings

Number of training cycles: 20

Use learned optimizer:

Learning rate: 0.0005

Training processor: CPU

Data augmentation:

Advanced training settings

Neural network architecture

Input layer (27,648 features)

Training output

(0)

Model

Model version: Quantized (int8)

Last training performance (validation set)

ACCURACY 33.3%

LOSS 1.25

Confusion matrix (validation set)

	FAN	KEYBOARI	MONITOR	MOUSE	NOTEBOO	PEN	WATCH
FAN	-	-	-	-	-	-	-
KEYBOARD	-	-	-	-	-	-	-
MONITOR	0%	0%	0%	0%	100%	0%	0%
MOUSE	-	-	-	-	-	-	-
NOTEBOOK	0%	0%	0%	0%	100%	0%	0%
PEN	-	-	-	-	-	-	-
WATCH	0%	0%	0%	100%	0%	0%	0%
F1 SCORE			0.00	0.00	0.67		0.00